

Architecture Improvements

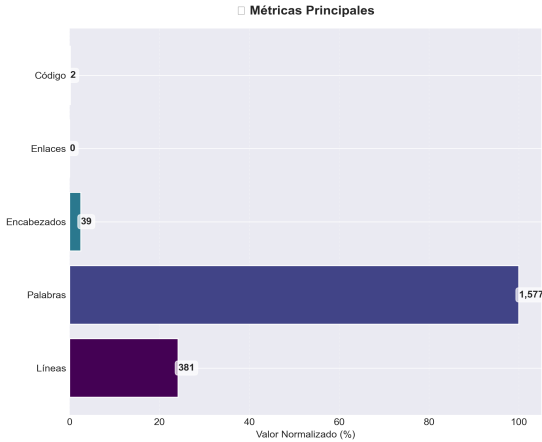
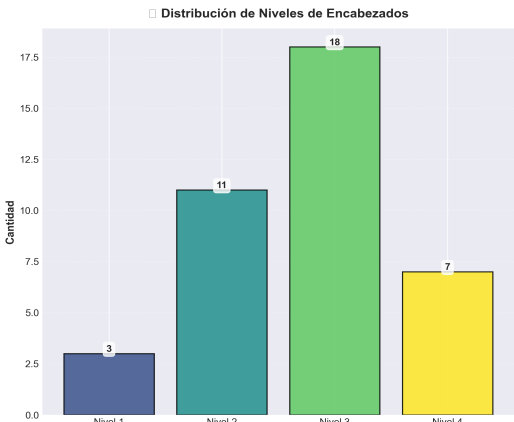
Análisis y Documentación Completa

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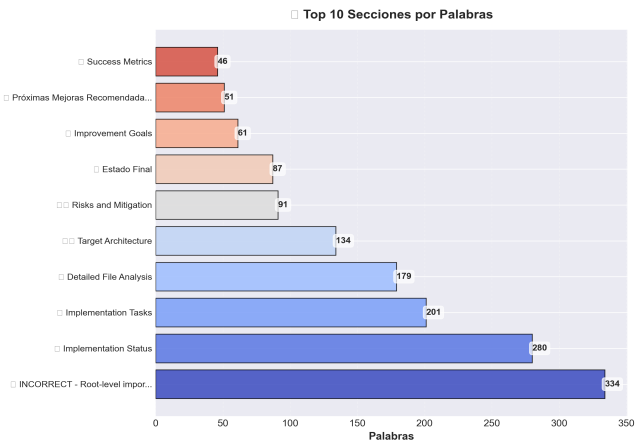
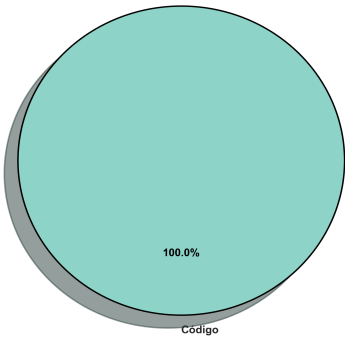
Análisis Visual Completo

Dashboard Principal

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 Análisis Completo: ARCHITECTURE_IMPROVEMENTS

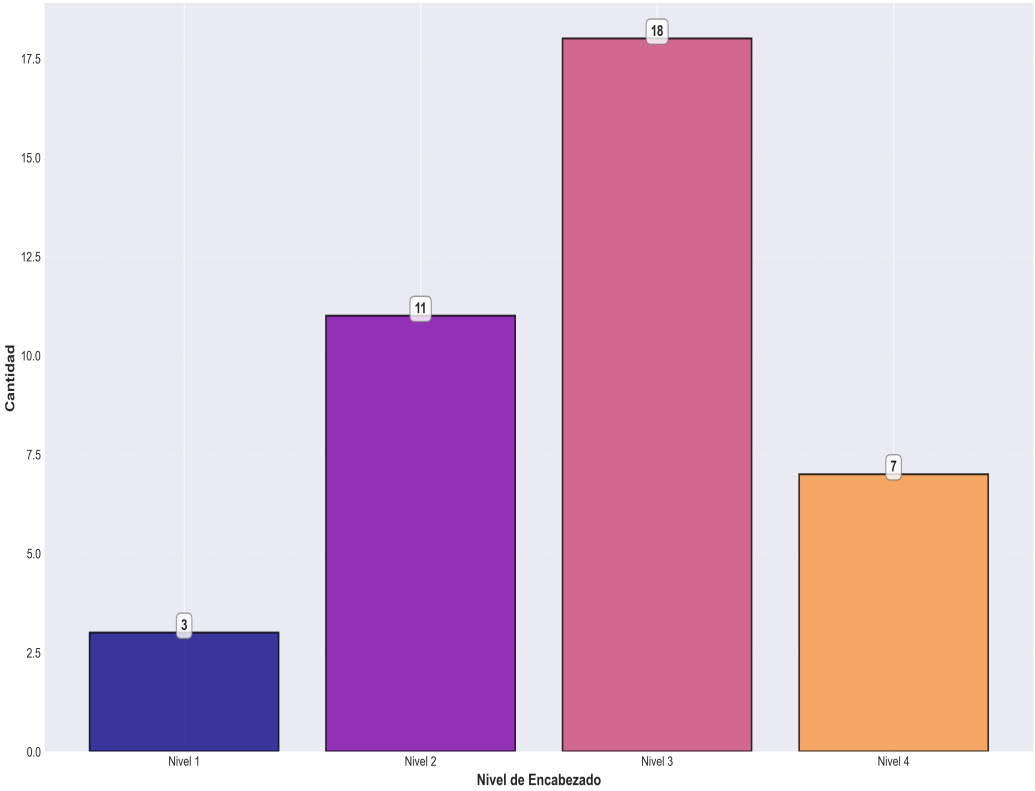


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 Distribución de Tipos de Contenido

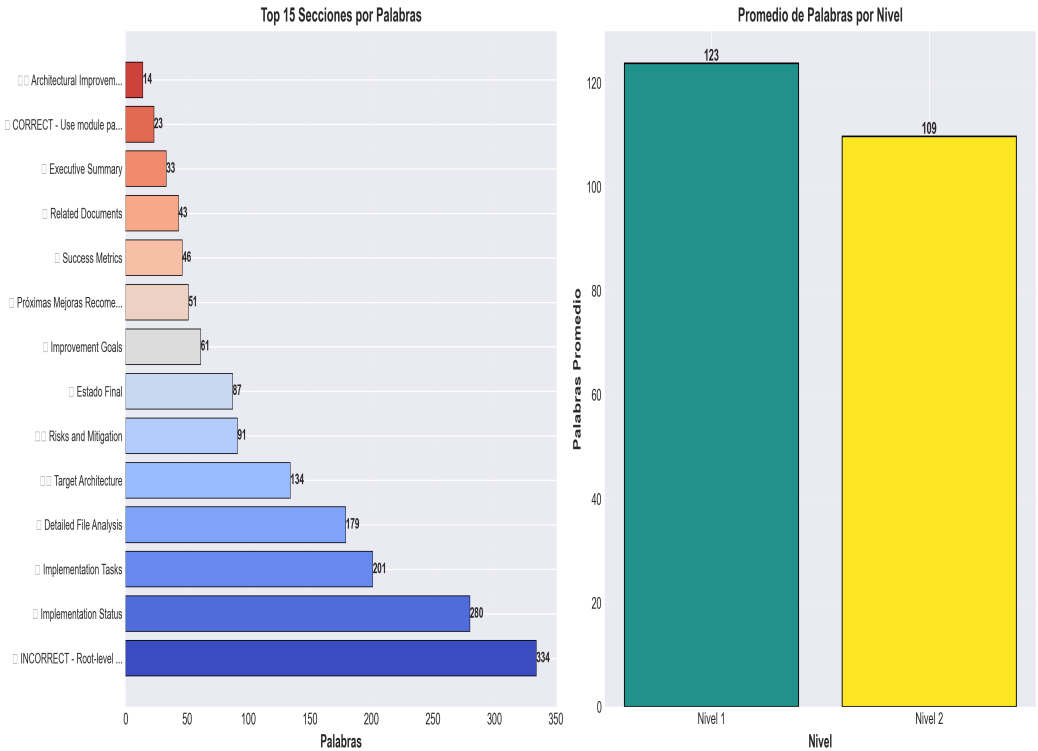


Análisis Adicionales

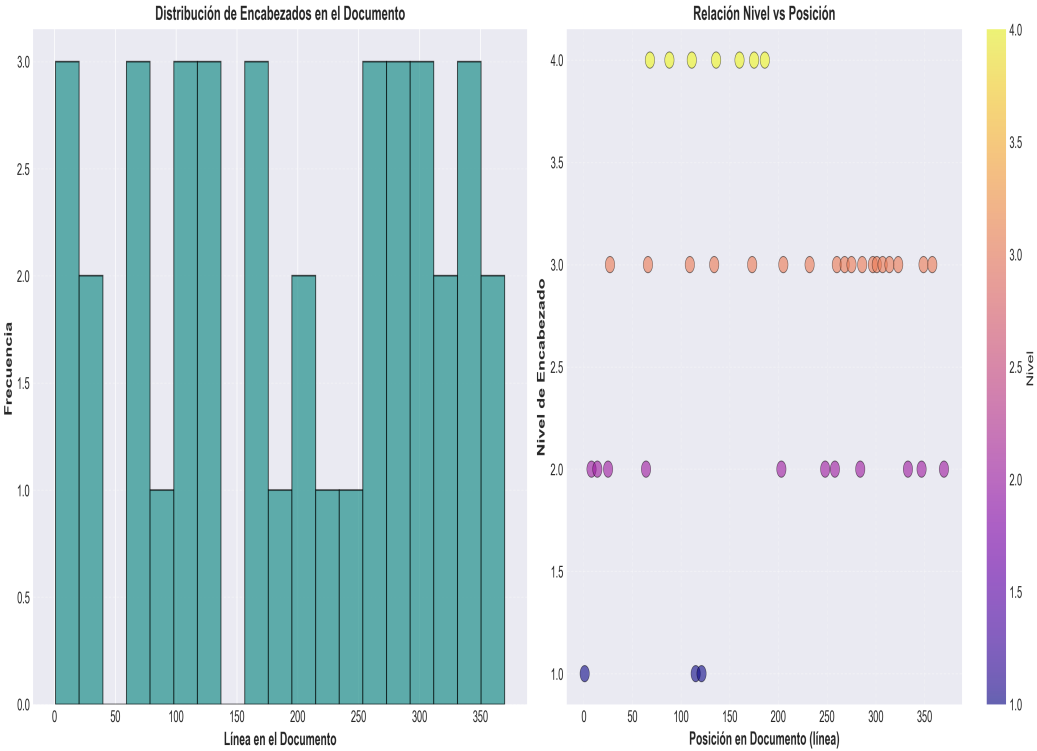
□ Estructura de Encabezados: ARCHITECTURE_IMPROVEMENTS



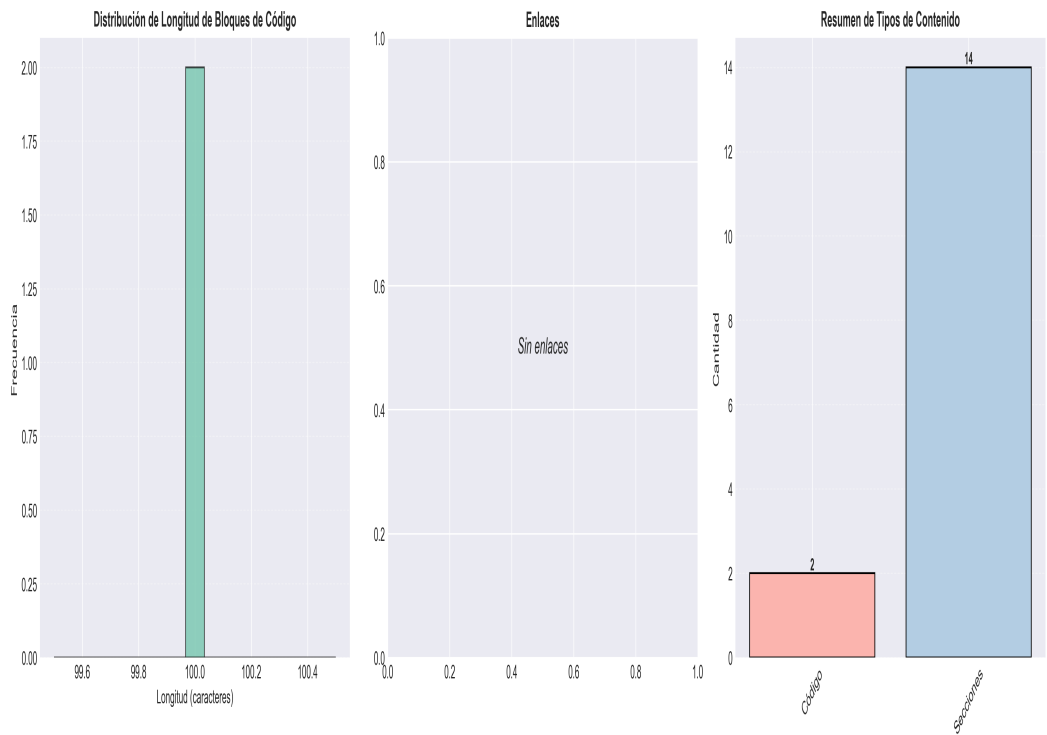
Análisis de Complejidad: ARCHITECTURE_IMPROVEMENTS



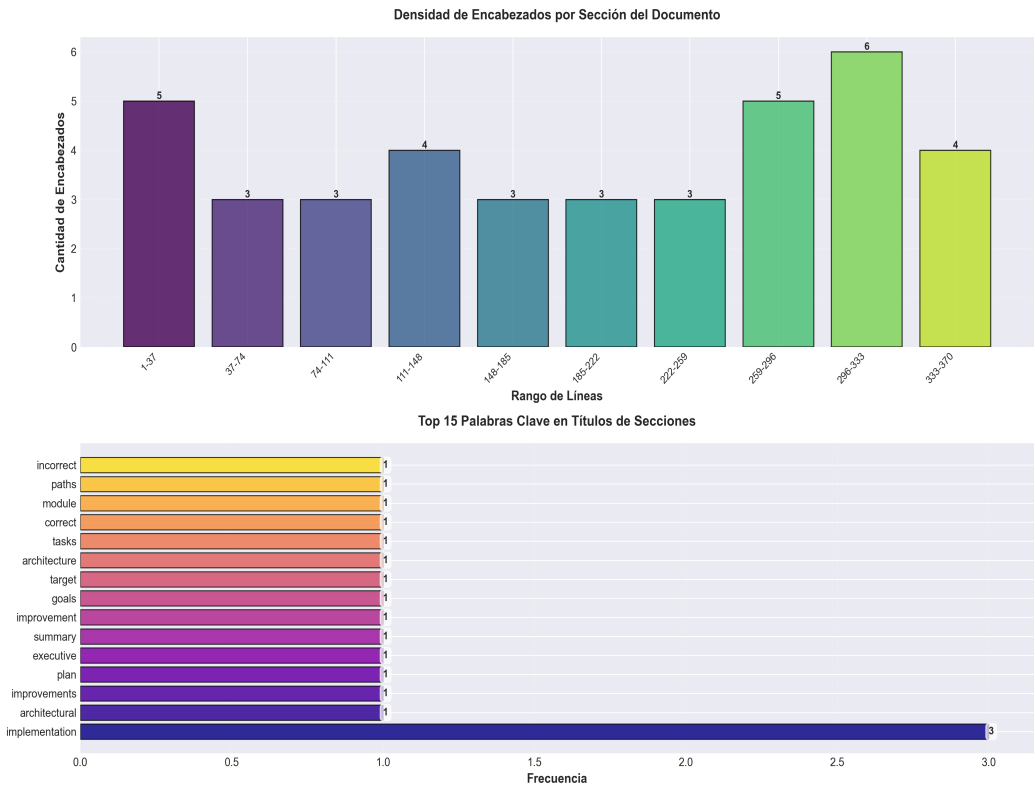
□
 Análisis Temporal y Distribución: ARCHITECTURE_IMPROVEMENTS



▣ Análisis Detallado de Contenido: ARCHITECTURE_IMPROVEMENTS



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 Análisis de Densidad y Frecuencia: ARCHITECTURE_IMPROVEMENTS



Resumen Ejecutivo: Este documento contiene 1,577 palabras distribuidas en 381 líneas, con 39 encabezados organizados en 14 secciones principales.

Estadísticas del Documento

Métrica	Valor
Total de líneas	381
Total de palabras	1,577
Total de caracteres	12,580
Encabezados	39
Bloques de código	2
Tablas	0
Enlaces	0
Imágenes	0
Secciones principales	14

Índice de Secciones

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Resumen del Contenido

■■ Architectural Improvements - Implementation Plan

****Date**:** 2025-01-27

****Status**:** Implementation In Progress

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■ Executive Summary

This document outlines the comprehensive architectural improvements being implemented to align the ``production_code`` directory with clean architecture principles, eliminate code duplication, standardize imports, and ensure proper layer separation.

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■ Improvement Goals

1. ****Eliminate Code Duplication****: Consolidate duplicate utility and config files
2. ****Standardize Imports****: Use consistent module paths throughout
3. ****Enforce Layer Separation****: Ensure proper dependency flow between layers
4. ****Resolve Naming Conflicts****: Fix Python built-in module conflicts
5. ****Improve Maintainability****: Clear separation of concerns and responsibilities
6. ****Enhance Testability****: Better structure for unit and integration testing

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■ ■ Target Architecture

Layer Structure

[REDACTED]
[REDACTED]

■ PRESENTATION LAYER ■

- - api/ (routes, middleware, auth, dependencies) ■

- - api_server.py, chat_server.py, cli*.py ■

- - dashboard.html ■

- Dependencies: application.service_container only ■

[illegible]

↓

[REDACTED]

[REDACTED]

■ APPLICATION LAYER ■

- - services/ (business logic services) ■

- - application/ (service container, orchestrators) ■

- - integration_pipeline.py, monitoring_system.py ■

- Dependencies: domain contracts, infrastructure providers■

[illegible]

↓

[REDACTED]

[REDACTED]

■ DOMAIN LAYER ■

- - core/ (base classes, utilities) ■

- - memory/, research/, inference/, etc. (domain models) ■

- - domain/contracts/ (interfaces, protocols) ■

- Dependencies: None (pure domain logic) ■

[illegible]

↓

[illegible]

■ INFRASTRUCTURE LAYER ■

- - infrastructure/providers/ (external integrations) ■

- - model data/, static/ (data access) ■

- Dependencies: domain contracts (implements them) ■

[REDACTED]

[REDACTED]

■ Implementation Tasks

Phase 1: File Consolidation ■

- ``api_utils.py`` (root, 268 lines) - Tensor validation functions
- ``core/api_utils.py`` (231 lines) - FastAPI/Flask helpers, HTTP clients
- ``api/api_utils.py`` (180 lines) - API route validation functions

1. ■ Keep ``api/api_utils.py`` for API route validation (domain-specific to presentation)
2. ■ Keep ``core/api_utils.py`` for general HTTP/web utilities (reusable infrastructure)
3. ■ Migrate tensor validation from root ``api_utils.py`` to ``api/api_utils.py``
4. ■ Update all imports
5. ■ Add deprecation warning to root ``api_utils.py``
6. ■ Remove root ``api_utils.py`` after migration

- `api\_unified.py` - Update imports
- `tests/test\_api\_utils.py` - Update imports
- Any other files importing from root `api\_utils`

- `config\_manager.py` (root, 521 lines)
- `core/config\_manager.py` (exists)

1. ■ Compare functionality between both files
2. ■ Consolidate into ``core/config_manager.py`` (infrastructure layer)
3. ■ Update all imports to use ``core.config_manager``

4. ■ Remove root `config\_manager.py` after migration

****Files to Update:****

[Nota: El documento original tiene 381 líneas. Se muestra un resumen de las primeras 100 líneas.]